

Project Proposal: Single Perceptron AI Hardware Accelerator

1. Introduction

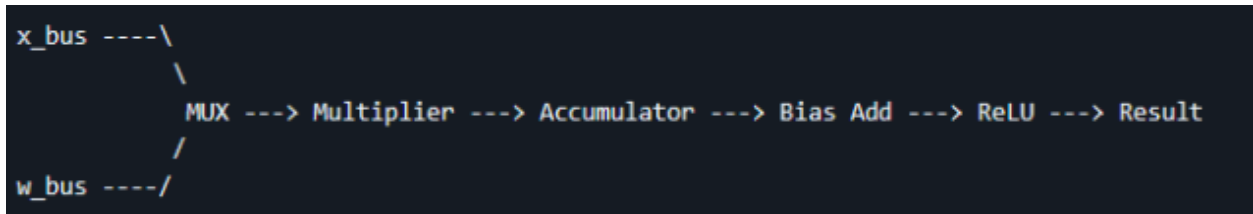
This project proposes the RTL design of a **Single Perceptron AI Hardware Accelerator**. The perceptron is the fundamental building block of Deep Neural Networks (DNNs), translating biological neuron mechanisms into a digital hardware datapath.



2. System Architecture & Specifications

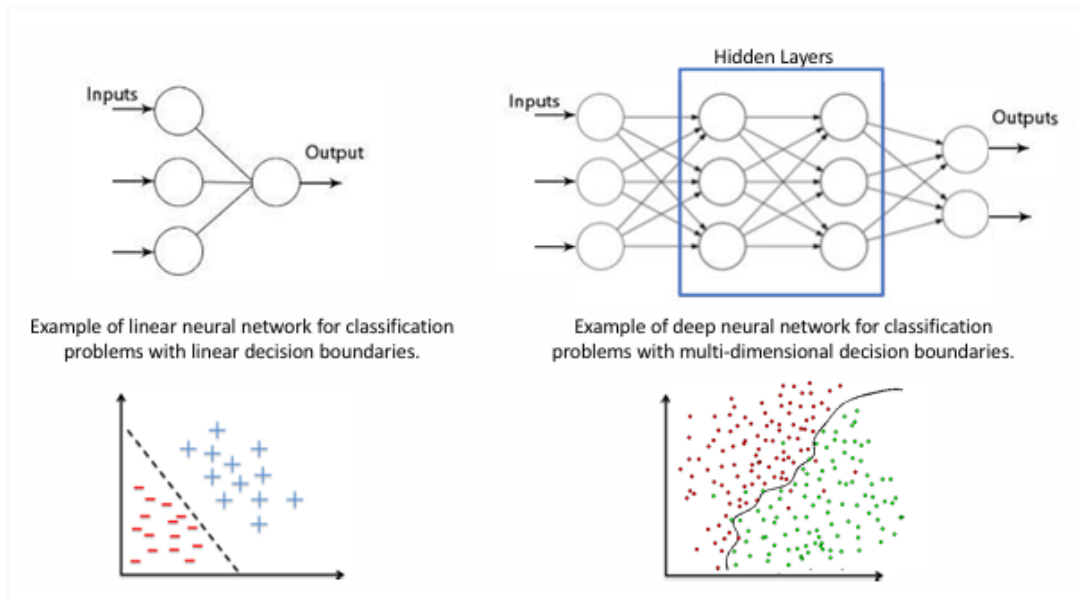
To optimize for embedded hardware constraints and real-time streaming, the design uses a Feed-Forward Datapath parameterized for different bit-widths. The default parameterization is set as follows:

- **parameter DATA_WIDTH = 8;** Minimizes area and latency by using 8-bit signed integers for inputs and weights instead of floating-point.
- **parameter ACC_WIDTH = 32;** Prevents data overflow during the Multiply-Accumulate (MAC) process.
- **ReLU Activation:** Replaces complex sigmoids with a simple MSB-check logic.



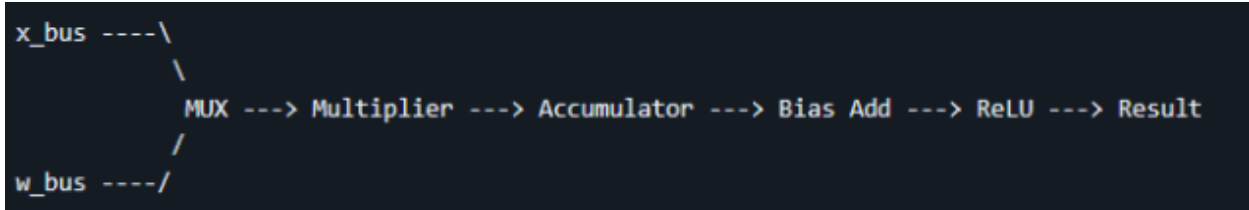
3. Hardware Datapath & Modules

The system is divided into parallel-developed modules. The datapath flows sequentially from the external inputs to the final decision.



Module	Description
1. MUX & Data Router	Selects the correct input data and weight based on the control signals.
2. Multiplier Unit	Combinational 8-bit x 8-bit multiplier generating a 16-bit product.

Module	Description
3. Accumulator Unit	32-bit clocked register storing the running sum (MAC Unit).
4. Bias Add Unit	Adds a parameterized 8-bit bias to the 32-bit accumulated sum.
5. ReLU Activation	Outputs 0 if the final sum is negative (MSB=1), else passes the sum.
6. FSM Controller	Manages data flow, Enables, and generates the 'done' signal.



4. Top-Level Interface (I/O)

The external interface accepts streaming data to reflect real-world AI applications where inputs

(e.g., sensor data or image pixels) change dynamically.

Port Name	Width	Direction	Description
clk	1	Input	System clock.
rst_n	1	Input	Active-low asynchronous reset.
start	1	Input	Trigger to begin computation.
data_valid	1	Input	Indicates the input data on the bus is valid.
x_in	[DATA_WIDTH-1:0]	Input	Streaming feature input (e.g., 8-bit).
w_in	[DATA_WIDTH-1:0]	Input	Streaming weight input (e.g., 8-bit).
final_output	[ACC_WIDTH-1:0]	Output	Decision value after ReLU (e.g., 32-bit).
done	1	Output	Flag indicating computation is complete.

5. Future Scope

This design serves as a scalable IP core. By instantiating multiple units in parallel, it can be expanded into a deep Convolutional Neural Network (CNN) integrated with a host processor via

an AXI-Stream interface for real-time inference.